Merjem Memic

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Education

University of Michigan — **Ann Arbor, MI**
B.S. in Data Science — *Expected May 2025*

- **GPA:** 3.85/4.00 | **Awards:** Dean's List, Academic Scholarship
- **Courses:** Data Structures & Algorithms, Practical Data Science, Web Systems, Machine Learning, Data Mining, Linear Regression, Statistics and Probability, Natural Language Processing (NLP), Computer Vision
- **Activities:** Muslim Engineering Society (Professional Development Chair), Michigan Data Science Team (Member)

Skills

Languages: Python, SQL, C++, R, HTML, CSS, JavaScript
Technologies: PyTorch, Pandas, Tableau, React, Node, Microsoft Excel, Firebase, AWS, Flask, Mongo DB
Interests: Hiking through the mountains and enjoying the open air, and crocheting small animals

Experience

University of Michigan — **Ann Arbor, Michigan**
Blue Lab Research Assistant — *Jan 2024 – Present*

- Performed sentiment analysis using the Bluesky Data API and NLP techniques to quantify a shift in attitudes compared.

Elementary Programming Concepts Course Grader — *Sept 2024 – Present*

- Evaluated coding projects for a class of 800+ students, ensuring alignment with industry standards.

Computer Science Engineering Mentor — *Sept 2024 – Present*

- Facilitated weekly mentoring sessions for a group of 4+ mentees per semester, addressing technical and professional topics, contributing to a program that supports 30+ students each semester in their journeys at university.

Headstarter AI — **Ann Arbor, Michigan**
Fellow — *Jul 2024 – Sept 2024*

- Built a basic Pantry Manager Application using Firebase, Next.js, Material UI, and React to allow control
- Programmed a Library Chatbot using the Gemini API on the backend and Next.js for the frontend

EcoData — **Ann Arbor Michigan**
Frontend Developer — *Jan 2024 – Apr 2024*

- Built an interactive page for a website to help individuals plan meals without excessive spending and food waste

Projects

Assessing Harmful Algal Blooms in Lake Erie

- Modeled multiple data sets of algae producing nutrients collected across Lake Erie using data visualization tools to realize an increase in the severity of blooms after COVID-19
- Utilized Python libraries Numpy, matplotlib, and scikit-learn to perform statistical analysis discovered a high correlation between nutrient content and water runoff
- Produced scatter plots, line plots, and heat maps with Python functions to visualize results to a non-technical audience and describe the future problems facing Lake Erie
- Cleaned data to ensure accuracy of observations and seamless integration into analysis

Instagram Clone

- Created responsive interfaces using React to implement authentication, post interactions, and personalized user accounts
- Applied Flask Python frameworks to develop REST APIs and route requests, and processed them by generating dynamic data from SQLite Database

Comparative Analysis of CNN, RNN, and SVM for Smartphone-Based Human Activity Recognition

- Implemented a Support Vector Machine, Convolution Neural Network, and Recurrent Neural Network to classify a person's activities based on sensor data collected from a smartphone's embedded accelerometer and gyroscope.

Factors in the Severity of Vehicle Crashes

- Modeled large data set of information on vehicle crashes which resulted in finding a daily trend in crash severity
- Modeled crash datasets using R to calculate the average severity of crashes for all seasons to detect a pattern of more severe crashes in the winter months on while more crashes overall happened in dry months

Euchre Interactive Card Game Interface

- Designed a C++ interface that allows users to play a game of euchre with simulated players executing moves via a programmed strategy through object-oriented programming